Isabella Pu

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RESEARCH OVERVIEW

My research explores **human-centered social robots** and **AI systems** that **support learning through creative expression**, using playful, interactive experiences that **blend AI with the arts**. I design, build, and evaluate these systems primarily in K-12 contexts, where creative activities—visual art, storytelling, music, games, and beyond—serve as powerful mechanisms for education and AI literacy. Central to my approach is **cultivating joy** and **delight**, with a goal of making interactions genuinely *fun*. Through my work, I enable users to understand subject matter and underlying principles of AI and robotics, empowering them to engage with these technologies safely, effectively, and intentionally. My work lies at the intersection of human-computer interaction (HCI), human-robot interaction (HRI), artificial intelligence (AI), and learning sciences.

EDUCATION

2025– **Ph.D.** in **Media Arts & Sciences**, Massachusetts Institute of Technology
Advised by Cynthia Breazeal

2023–25 **M.S.** in **Media Arts & Sciences**, Massachusetts Institute of Technology
Advised by Cynthia Breazeal
Thesis: *Interactive Storybooks for Early AI Literacy*
Committee Members: Cynthia Breazeal, Scot Osterweil, Joe Blatt

2019–23 **B.S.E** in **Computer Science**, Princeton University, *magna cum laude*
Certificate in **Robots & Intelligent Systems**
Advised by Naomi Leonard, Jeffrey Snyder, and Radhika Nagpal
Thesis: *A Robotic Generative Percussion Quartet*

PUBLICATIONS

Conference Proceedings

[1] Who's the Boss? Children Negotiate Robot Control Across Role and Context
Isabella Pu, Kantwon Rogers, Linh Dieu Dinh, Sharifa Alghowinem, Cynthia Breazeal
Accepted to HRI 2026.

[2] "How can we learn and use AI at the same time?": Participatory Design of GenAI with High School Students
Isabella Pu, Prerna Ravi, Linh Dieu Dinh, Chelsea Joe, Caitlin Ogoe, Zixuan Li, Cynthia Breazeal, Anastasia K. Ostrowski.
ACM IDC 2025

[3] The *Beatbots*: A Musician-Informed Multi-Robot Percussion Quartet
Isabella Pu, Jeff Snyder, Naomi Ehrich Leonard.
ACM/IEEE HRI 2025

[4] A HeARTfelt Robot: Social Robot-Driven Deep Emotional Art Reflection with Children
Isabella Pu, Golda Nguyen, Lama Alsultan, Rosalind Picard, Cynthia Breazeal, Sharifa Alghowinem.
IEEE RO-MAN 2024
Winner of *RSJ Pioneering Research Award in Robot and Human Interactive Communication*

FELLOWSHIPS

2025 MIT Presidential Graduate Fellowship

AWARDS AND HONORS

2025 IEEE Robots & Automation Society Travel Grant, ACM/IEEE HRI 2025 ($1500

2024 RSJ Pioneering Research Award in Robot and Human Interactive Communication, IEEE RO-MAN 2024

2023 Outstanding Undergraduate Thesis in Computer Science, Princeton University

2023 Sigma Xi Membership, Princeton University

2022 Center on Science and Technology Research Grant, Princeton University

INVITED TALKS

2025 **AI and Robots for Fun and Engaging K-12 Education**
Invited talk at Robert A. Van Wyck M.S. 217Q (NYC Public Schools)

2025 **Intro to Child-AI and Child-Robot Research**
Invited talk at *Expedition AI: Demystify Artificial Intelligence* (American Museum of Natural History)

2025 **Child-Centered AI and Robots for Effective + Engaging Education**
Guest lecture for *Designing and Producing Media for Learning*, Joe Blatt (Harvard Graduate School of Education)

2025 **Robots, AI, and the Future of Education**
Guest lecture for *How the Future of Work is Shaping the Future of Education*, Prof. Peter Quartermaine Blair (Harvard Graduate School of Education)

2023 **Multi-Robot Generative Percussion Music**
Guest lecture for *COS IW Seminar — Swarm Intelligence: Ants, Bees, Fish, and Humans*, Prof. Radhika Nagpal (Princeton University)

ART

2022 Featured animations on Generative Adversarial Networks (GANs) in *The Imitation Game: Visual Culture in the Age of Artificial Intelligence*, Vancouver Art Gallery

RESEARCH PROJECTS

2023– **Interactive Storybooks for Early AI Literacy**

Advisor: Cynthia Breazeal

Leading an interdisciplinary team to build a suite of Svelte-based interactive storybooks that teach K-3 students foundational AI and robotics concepts through fun narrative, games, and AI interactions in creative contexts. The system integrates multimodal generative AI—including text, image, and music generation—within a scaffolded framework for child-safe interaction. Each storybook includes a custom UI optimized for early learners and AI-powered mini-games that teach concepts such as model training, robot perception, and pattern recognition through playful, multi-modal AI interaction.

2023– ***Doodlebot:* Educational Robotics System + Curricula for Creative AI Education**

Advisor: Cynthia Breazeal

Collaborating with designers and engineers on Doodlebot, an affordable classroom social robot combining Arduino and Raspberry Pi. Doodlebot features a bespoke drag-and-drop coding interface for movement, drawing, social interaction, sensor processing, and real-time AI models. Developed a companion middle-school curriculum on programming and AI concepts such as model training and bias, and conducted studies in which K-2 children children programmed Doodlebot and explored preferences on robot control in different settings, aesthetic trait and role expression, and social friction with robots.

2023– **Social-Emotional Learning with Robots via Arts Methodologies**

Advisors: Rosalind Picard, Cynthia Breazeal, Sharifa Alghowinem

Designed and developed fully autonomous child-robot system for children aged 7-11 to discuss art with a social robot to practice social-emotional skills. The Python-based system utilizes generative AI, enabling the robot to produce context-sensitive dialogue that responds adaptively and references earlier statements.

Developing a child-robot platform for children aged 5-8 that facilitates emotional storytelling via a social robot and physical props. Combines CNN-based vision models and vision-language models to reference props, while integrating generative AI for safe, dynamic, and context-sensitive dialogue and storytelling.

2023–24 **Collaborative Storybook Creation with AI for Elementary School**

Advisors: Safinah Ali, Cynthia Breazeal

Developed a full Unity-based child-AI storybook creation system integrating generative AI APIs. Designed a child-friendly tablet interface and scaffolded AI interactions to support safe, age-appropriate use and encourage divergent thinking and complex storytelling.

2023–24 **Climate Data Exploration & Visualization Curriculum for High School**

Advisors: Matthew Taylor, Cynthia Breazeal

Created high school CS curriculum and educator guide to teach how Python can be used to explore and visualize climate data. Curriculum included data cleaning using *pandas* and *numpy*, linear regressions and SVMs in *scikit-learn* as predictive climate models, data visualization using *matplotlib* and *seaborn*, and modules on data theatre. Published online to students in over 114 countries and all 50 U.S. states.

2022–23 **Generating Percussion Music with a Multi-Agent Robotic System**

Advisors: Naomi Leonard, Jeffrey Snyder, Radhika Nagpal

Designed and developed a robotic percussion quartet using Sphero BOLT robots in a musician-informed design process which played generative music using Python. Audience members interacted with robots by picking them up, adding obstacles, and playing a MIDI keyboard to change robots' colors and movements.

2021–22 **Educational GANs Animation for Vancouver Art Gallery**

Advisor: Adam Finkelstein

Designed and coded two animations in p5.js targeting non-technical audiences, teaching viewers about training GANs. Trained and optimized GANs to generate numerical digits and cats. Animations displayed in *The Imitation Game* exhibit at the Vancouver Art Gallery from 03/05/2022 - 10/23/2022.

PROFESSIONAL EXPERIENCE

2021–23 **Software Engineering Summer Intern**, Blizzard Entertainment (Player Interactions & Trust). Irvine, CA.
Projects included developing neural networks to predict player churn based on disruptive behavior and building data pipelines for an industry-first positive player behavior analysis platform—implemented in multiplayer online games with millions of players (Overwatch, World of Warcraft)

2020–23 **Undergraduate Course Assistant/Grader**, Princeton University. Princeton, NJ.
Fall 2020: *Introduction to Programming Systems*
Spring 2021: *Introduction to Machine Learning*
Fall 2021: *Computer Vision*
Spring 2022 and 2023: *Economics and Computation*

2016–19 **Kid Code Teacher**, The Westminster Schools. Atlanta, GA.
Taught Scratch and basic Python coding to 2nd through 5th graders at after-school program.

SERVICE

Service to the University

2024– MIT Graduate Residence Advisor, Fariborz Maseeh Hall

Reviewer

ACM Conference on Designing Interactive Systems (DIS)

ACM Conference on Human Factors in Computing Systems (CHI)

ACM/IEEE Conference on Human-Robot Interaction (HRI)

International Journal of Child-Computer Interaction (IJCCI)

Mentorship

2024– Linh Dieu Dinh, *Wellesley College*

2023– Jimmy Kuhlman, *MIT*

2025 Megan Yi, *Wellesley College*

2024-25 Si Liang Lei, *MIT*

2024-25 Adwa Alghihab, *Prince Sultan University*

2024-25 Sara Al Mogren, *Prince Sultan University*

2024 Hyun Kim, *MIT*

2024 Joanne Hong, *MIT*

2024 Alessandro Briseño-Tapia, *MIT*

2023 Laura Morris, *MIT*

Leadership

2020–23 *President*, Princeton Women in Computer Science. Princeton, NJ.

2020–22 *FIRST Robotics Competition Mentor*, Robbinsville High School. Robbinsville, NJ.

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